CLOUD EDUCATION LEARNING (P) LTD
Registered Address: K No- 2121041180X, Chakkarpur, Gurugram, Haryana -122002
Branch Address: D-12 B 1st Floor, Chhatarpur Enclave, 100 Feet Road, New Delhi-110074
Contact: 011-43634116
Email:info@cloudeducation.co.in
Website: www.cloudeducation.co.in

Proposal for 3-Day Training Program

Machine Learning using Python

Submitted By
Cloud Education Learning Pvt. Ltd.
Brand: **AiBo8**

Program Overview
Cloud Education Learning Pvt. Ltd. proposes a **3-Day Training Program on Machine Learning using Python** designed for **university students (beginner to intermediate level)**. The objective of this program is to provide students with **practical exposure to Python programming, data analysis, and machine learning concepts** through hands-on exercises and real-world datasets.
The training program will focus on **conceptual understanding along with practical implementation**, enabling students to build simple machine learning models by the end of the program.

Program Details
Program Title: Machine Learning using Python
Duration: 3 Days
Daily Training Duration: 5 Hours
Session Structure

- **Session 1:** 2.5 Hours (Before Lunch)
- **Session 2:** 2.5 Hours (After Lunch)

Target Audience:
University / Engineering students interested in AI, Data Science, and Machine Learning.
Prerequisite:
Basic programming knowledge is helpful but **not mandatory**.

Day 1: Python Programming for Data Science
Objective
To build a strong foundation in **Python programming** required for data science and machine learning.

Session 1 (2.5 Hours): Python Fundamentals

1. Introduction to Python

- Why Python for AI and Machine Learning
- Python ecosystem for Data Science
- Installing Python environment
- Introduction to **Jupyter Notebook / Google Colab**
- Writing and executing the first Python program

2. Python Basics

- Variables and Data Types
- Integer, Float, String, Boolean
- Type Conversion

3. Python Operators

- Arithmetic Operators
- Comparison Operators
- Logical Operators
- Assignment Operators

4. Control Structures

- Conditional Statements
- if statements
- if–else statements
- Nested conditions

5. Loops

- for loop
- while loop
- break and continue

Hands-on Activity

Students will write small Python programs such as:

- Even/Odd Number Checker
- Prime Number Checker
- Multiplication Table Generator

Session 2 (2.5 Hours): Python for Data Handling

6. Python Data Structures

- Lists
- Tuples
- Sets
- Dictionaries

7. Functions in Python

- Defining functions
- Function arguments
- Return statements
- Lambda functions

8. File Handling
- Reading text files
- Writing files
- Basics of CSV files

9. Introduction to Python Libraries
- What are Python libraries
- Installing libraries using pip
- Overview of Data Science libraries

Mini Practice
Students will implement:
- Student record system using dictionary
- List operations and sorting
- Simple file read/write program

Day 2: Data Analysis using Python

Objective
To teach students how to **analyze, process, and visualize data using Python libraries**.

Libraries Covered
- NumPy
- Pandas
- Matplotlib
- Seaborn

Session 1 (2.5 Hours): Numerical Computing & Data Handling

1. Introduction to Data Analysis
- Data Science workflow
- Structured vs Unstructured Data
- Dataset formats: CSV, Excel, JSON

2. NumPy (Numerical Python)
Concepts Covered:
- NumPy arrays
- Array creation
- Indexing and slicing
- Array operations
- Mathematical functions
- Broadcasting

Practical Exercises
- Creating arrays
- Matrix operations
- Statistical calculations

3. Pandas (Data Analysis Library)

Concepts Covered:

- Series
- DataFrame
- Importing datasets
- Viewing and inspecting data
- Handling missing values
- Filtering and sorting
- GroupBy operations

Practical Exercises

- Load CSV dataset
- Perform data cleaning
- Data selection and manipulation

Session 2 (2.5 Hours): Data Visualization

4. Data Visualization Basics

- Importance of visualization in data analysis

5. Matplotlib

Concepts Covered:

- Line plots
- Bar charts
- Scatter plots
- Histograms
- Plot customization

Practical Exercises

- Plotting dataset trends
- Visualizing statistical distributions

6. Seaborn (Advanced Visualization)

Concepts Covered:

- Statistical plots
- Heatmaps
- Pairplots
- Correlation visualization

Practical Exercises

- Dataset correlation analysis
- Feature visualization

Mini Project

Students will perform:

- Dataset exploration
- Data cleaning
- Data visualization

Example datasets:

- Student Performance Dataset
- Sales Dataset

Day 3: Machine Learning with Python

Objective

To introduce students to **Machine Learning concepts and implement ML models using Python**.

Libraries Used

- Scikit-learn
- Pandas
- NumPy

Session 1 (2.5 Hours): Machine Learning Fundamentals

1. Introduction to Machine Learning

- What is Machine Learning
- Difference between AI, ML, and Deep Learning

Types of Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

2. Machine Learning Workflow

- Dataset collection
- Data preprocessing
- Feature selection
- Model training
- Model evaluation

3. Data Preprocessing

- Train/Test split
- Feature scaling
- Handling missing data
- Encoding categorical variables

4. Regression Algorithms

Concepts Covered:

- What is Regression
- Linear Regression

Practical Implementation

Students will build models to:

- Predict house prices
- Predict student marks

Steps involved:

- Load dataset
- Split training and testing data
- Train model
- Predict results
- Evaluate model

Session 2 (2.5 Hours): Classification Algorithms

5. Classification in Machine Learning

- What is classification
- Classification vs Regression

6. Classification Algorithms

Algorithms Covered:

- Logistic Regression
- Decision Tree
- K-Nearest Neighbors (KNN)

7. Model Evaluation

Concepts Covered:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1 Score

8. Mini Machine Learning Project

Students will build a **complete ML model**.
Example projects:

- Iris Flower Classification
- Spam Email Detection
- Student Pass/Fail Prediction

Steps performed:

- Load dataset
- Data preprocessing
- Train ML model
- Model evaluation
- Predictions

Final Outcome of Training

After completing this **3-day training program**, students will be able to:

- Write Python programs confidently
- Perform data analysis using **NumPy and Pandas**
- Create visualizations using **Matplotlib and Seaborn**
- Understand the **Machine Learning workflow**
- Implement **Regression and Classification models**
- Build simple **Machine Learning projects**

Commercials

Training Program Fee:
₹ 15,000 + Applicable Taxes
Payment Terms:
100% payment to be made prior to the commencement of the training program.